

GIWON SONG

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EDUCATION

Seoul National University

Bachelor of Science in Electronic and Computer Engineering 2029 (expected)

- Cumulative GPA: 4.04 / 4.3
- Academic Excellence Scholarship (Merit-based) : 2023 – 2024

University of California, Berkeley

Physics Major (Exchange Program) Jan 2027 – May 2027

Korea Science Academy of KAIST (KSA)

High School Diploma 2023

RESEARCH EXPERIENCE

Republic of Korea Army (ROKA)

Cryptanalyst May 2025 – Nov 2026

- Decrypted complex ciphers by training a large language model (LLM) targeted to specific properties of the plaintexts.

Quantum Information and Quantum Computer Lab (QuiQCL), Seoul National University

Undergraduate Research Intern July 2024 – Apr 2025

- Conducted research on the theoretical limits of quantum computing and algorithmic efficiency.

Undergraduate Research Opportunity Program (UROP) Participant July 2023 – Aug 2023

- Co-developed IQUIP (Ion trap Quantum computing User Interface Project). Project Repository : [\[GitHub\]](#)

Korea Science Academy of KAIST (KSA)

Student Researcher 2021

- Researched advanced implementations of Neural Radiance Fields (NeRF) for multiple objects.

PUBLICATIONS

Published / Pre-prints

- Hwang, Y.^{*}, Song, G.^{*}, Park, K., Kwon, H.[†], Kim, T.[†] “Efficient expectation value estimation for quantum circuits via extended stabilizer frameworks and adaptive variance estimation.” ArXiv preprint, 2026. ArXiv : [\[ArXiv\]](#)
- Park, K., Choi, K., Kim, M., Song, G., Kim, T.[†] “Quantum Linear Multistep Method for Using a Quantum Oracle with Differential Equations.” ArXiv preprint, 2025. ArXiv : [\[ArXiv\]](#)

^{*}These authors contributed equally to the work. [†]Corresponding author(s).

AWARDS & HONORS

- Special Prize (IBM Quantum) | Korea Q Hackathon 2024
- Advanced Badge (Completed all exercises) | IBM Quantum Spring Challenge 2024
- Director's Award, Quantum Information Research Support Center | Korea Q Hackathon 2023
- Silver Prize | Korea Olympiad in Informatics (KOI) 2016

SKILLS & LANGUAGES

Technical skills: C++, Python, Qiskit, LaTeX, Git

Languages: Korean (native), English (professional working proficiency), Japanese (professional working proficiency)

- iBT TOEFL: 5.5/6 (110/120)
- JLPT: N1 (127/180), N2 (165/180)